

# MTH303 - Midsem 1

## The FL Axioms:

1.  $p \rightarrow (q \rightarrow p)$
2.  $[p \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$
3.  $\neg p \rightarrow \neg q \rightarrow (q \rightarrow p)$

1. Consider the two expressions  $P \rightarrow (P \vee Q)$ ,  $P \rightarrow (P \wedge Q)$ .
  - (a) (2 points) Which of these two expressions are tautologies? Why?
  - (b) (2 points) Is it possible to write each of these two expressions entirely in terms of  $\rightarrow$  and/or  $\neg$ ? Explain.
  - (c) (3 points) Can you prove each of them using only the axioms and the replacement rules? If a proof exists, give one; if one does not exist, explain why. (Context: We are allowed the FL Axioms, Replacement Rules, and Modus Ponens)
2. (2 points) Prove that if  $\vdash p \rightarrow q$ , then,  $p \vdash q$  for all propositional forms  $p, q$ , i.e. if there is a proof of  $p \rightarrow q$ , then, there is a proof of  $q$  assuming the hypothesis  $p$ . Explain clearly and completely.
3. (3 points) Why does soundness imply consistency in propositional logic?
4. (2 points) Consider a given sequence of propositional forms  $p_0, p_1, \dots$ . Suppose there is a propositional form  $q$  for which there is no proof of  $q$  from the given sequence, i.e.  $p_0, p_1, \dots, \not\vdash q$ . Then, prove that the given sequence is consistent. Recall that  $p_0, p_1, \dots$  are inconsistent if  $p_0, p_1, \dots, \vdash r \wedge \neg r$  for some propositional form  $r$ , and they are consistent if they are not inconsistent.
5. Consider a first order language consisting of one relational/predicate symbol,  $\in$ , and one constant symbol,  $\emptyset$ . Let us consider the structure  $\mathcal{N}$ , where  $|\mathcal{N}| = \{0, 1, 2, \dots\}$ , i.e. the usual natural numbers, and 0. Define  $\emptyset^{\mathcal{N}} = 0$  and  $\in^{\mathcal{N}} = <$ . (Context: the  $\in$  relation is meant to be the usual  $<$  relation on  $\mathbb{N}$ )
  - (a) (2 points) Is  $\forall x (\emptyset \in x)$  true in  $\mathcal{N}$ ?
  - (b) (2 points) Is  $\forall x (\neg(x \in \emptyset))$  true in  $\mathcal{N}$ ?
  - (c) (2 points) Is it true that  $\models \forall x (\neg(x \in \emptyset))$ ? Recall that  $\models \phi$  means  $\emptyset \models \phi$  and  $\Gamma \models \phi$  means " $\Gamma$  logically implies  $\phi$ ".